

Term Project Option 1

Adaptive Color Plane Interpolation

1. Introduction

In this project, you are asked to develop a color interpolation accelerator for digital still cameras (DSC). In single CCD camera systems, images are captured in a Bayer CFA format as shown in Figure 1. Demosaicing or color interpolation is the process to recover a full-color image from the mosaic image. In DSCs, this interpolation process must be performed in real time and a hardware accelerator helps speed up the operation.

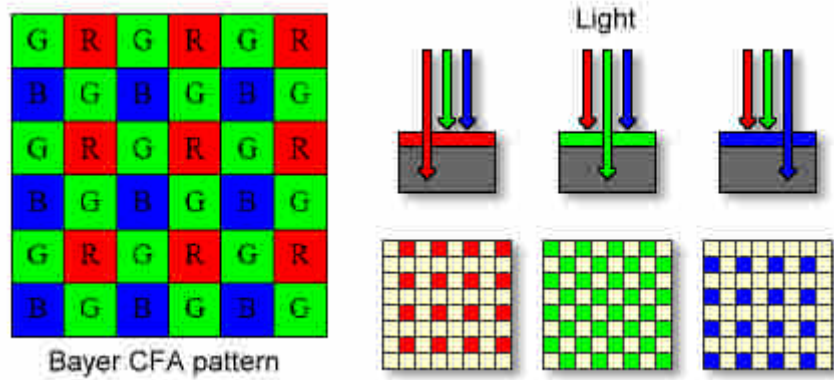


Figure 1. Bayer CFA patterns captured by single CCD cameras

2. Adaptive color plane interpolation (ACPI) scheme

Refer to Figure 2, the interpolation scheme used in this project is ACPI and the scheme can be described by the following procedures:

- Green interpolation

- Form two classifiers

$$DH = |G_6 - G_8| + |2R_7 - R_5 - R_9|$$

$$DV = |G_3 - G_{11}| + |2R_7 - R_1 - R_{13}|$$

- IF $DH < DV$

$$\hat{G}_7 = (G_6 + G_8) / 2 + (2R_7 - R_5 - R_9) / 4$$

- ELSE IF $DV < DH$

$$\hat{G}_7 = (G_3 + G_{11}) / 2 + (2R_7 - R_1 - R_{13}) / 4$$

- ELSE

$$\hat{G}_7 = (G_3 + G_{11} + G_6 + G_8) / 4 + (4R_7 - R_1 - R_{13} - R_5 - R_9) / 8$$

- Blue implementation

- Interpolate B_3, B_6, B_8 and B_{11}

$$\hat{B}_3 = (B_2 + B_4) / 2 + (2G_3 - \hat{G}_2 - \hat{G}_4) / 2$$

$$\hat{B}_6 = (B_2 + B_{10}) / 2 + (2G_6 - \hat{G}_2 - \hat{G}_{10}) / 2$$

- Interpolate B_7

- ✓ Form two classifiers

$$DN = |B_2 - B_{12}| + |2\hat{G}_7 - \hat{G}_2 - \hat{G}_{12}|$$

$$DP = |B_4 - B_{10}| + |2\hat{G}_7 - \hat{G}_4 - \hat{G}_{10}|$$

- ✓ IF $DN < DP$

$$\hat{B}_7 = (B_2 + B_{12}) / 2 + (2\hat{G}_7 - \hat{G}_2 - \hat{G}_{12}) / 2$$

- ✓ ELSE IF $DP < DN$

$$\hat{B}_7 = (B_4 + B_{10}) / 2 + (2\hat{G}_7 - \hat{G}_4 - \hat{G}_{10}) / 2$$

- ✓ ELSE

$$\hat{B}_7 = (B_4 + B_{10} + B_2 + B_{12}) / 4 + (4\hat{G}_7 - \hat{G}_4 - \hat{G}_{10} - \hat{G}_2 - \hat{G}_{12}) / 4$$

- Red implementation can be obtained in a similar way

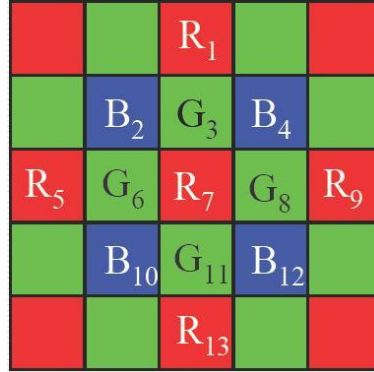


Figure 2. pixel labeling

3. Hardware platform and design guidelines

As shown in Figure 3, the target hardware platform consists of one hardware accelerator and two memory modules. Memory module 1 contains the original Bayer CFA images. It is a single port memory and has a data width equal to 8-bit. (Note that each color element of a pixel is 8-bit wide) The data layout in module 1 follows the raster scan order of the Bayer CFA pattern. Memory module 2 is also a single port memory and has a data width of 24-bit. It is used to store the interpolated image and the format should be the (RGB) in a raster scan order. Inside the hardware accelerator, you are allowed to implement line buffers or register files if necessary. The hardware accelerator is assumed to access data from memory module 1 in a constant data rate and to write the interpolated data to memory module 2 at

the same pixel rate. Ideally, you should access each data pixel from memory module 1 only once and process at a throughput 1 pixel/clock. If your design cannot meet this ultimate goal, you may have multiple accesses and use more than one clock cycle to process one pixel. A sample .bmp image containing all RGB components and its corresponding Bayer CFA image will be available at the course web site later to serve as the test bench of your design. There are two distortion measurement indexes:

- $$MSE = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W (f(i, j) - \hat{f}(i, j))^2$$
- $$PSNR(x, y) = 10 \log_{10} \frac{255^2}{MSE(x, y)}$$

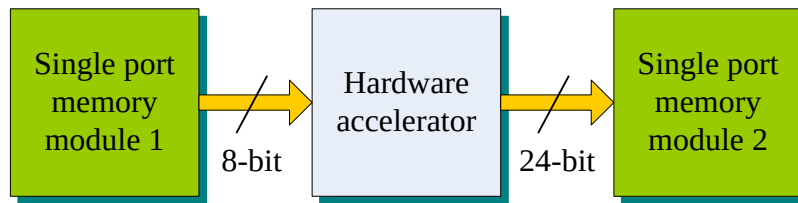


Figure 3. Target hardware platform

4. Design Merits

The effectiveness of your design will be evaluated based on the following performance indexes:

- Total logic gate count – for the data path and the control path
- Total on-chip memory size – please provide your memory configurations and the equivalent gate count as well
- Degree of computing parallelism – the number of concurrent pixel predictions
- Internal memory access bandwidth – in the unit of MB/sec
- External memory access bandwidth – in the unit of MB/sec
- Maximum working frequency – in the unit of MHz
- Clock cycles needed to complete the prediction of one image
- Throughput rate – number of images can be processed per second
- Power consumption at maximum working frequency
- Energy needed to complete the prediction of one image = power/throughput rate

5. Hand in requirements

- A report in MS word format highlights your architecture design, data path design, memory configuration, simulation and synthesis results, and the performance indexes stated above. You are also suggested to describe the algorithm mapping techniques applied in this project.
- A power point file to give an overview of your design.

- A tar file containing all design files and documents for the uploading purpose.